GROUP -7 DOCUMENTATION
MILESTONE - 4 Dashboards and Deployment

1. Objective

The objective of this milestone is to design and develop an interactive dashboard using Streamlit, integrate it with the ETL pipeline built using Apache Airflow, validate the processed data, and prepare the system for deployment.

2. System Overview

The system follows a complete data engineering workflow:

- Raw CSV data is ingested into the system
- Apache Airflow performs ETL (Extract, Transform, Load) operations
- Cleaned data is stored in a SQLite database (`infosys.db`)
- The Streamlit dashboard fetches and visualizes the processed data

This pipeline ensures efficient data processing and real-time analytics.

3. Tools and Technologies Used

Tool	Purpose
Apache Airflow	ETL pipeline orchestration
SQLite	Database storage
Streamlit	Dashboard development
Pandas	Data processing and merging
Matplotlib	Data visualization

4. Database and Tables Used

The dashboard uses the following cleaned tables:

• clean_orders

- order_id

- order_date
- customer_name
- state
- city

• **clean_order_details**

- category
- sub_category
- quantity
- amount
- profit

• **clean_sales_target**

- month
- category
- target

5. Data Processing and Merging

Data from multiple tables is merged using **Pandas** to create a unified dataset.

- Merging is performed using the **order_id** key
- Combines order-level and product-level data
- Enables multi-dimensional analysis (time, category, customer, etc.)

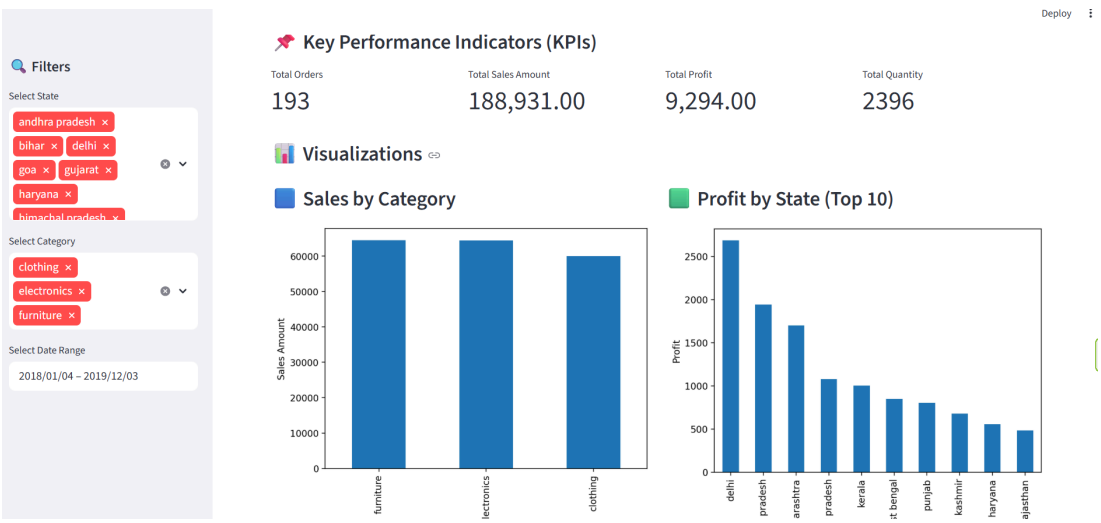
This step is crucial for generating meaningful insights in the dashboard.

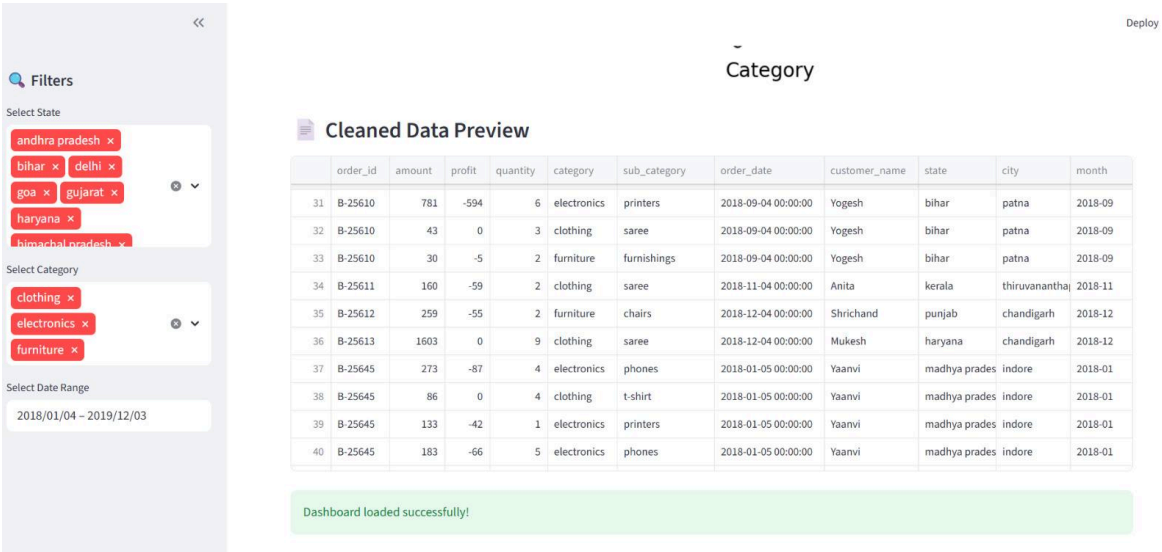
6. Dashboard Development

An interactive dashboard was developed using **Streamlit** to analyze cleaned ETL data.

Key Features:

- Interactive filters
- Real-time updates
- Visual analytics
- Data preview table





	order_id	amount	profit	quantity	category	sub_category	order_date	customer_name	state	city	month
31	B-25610	781	-594	6	electronics	printers	2018-09-04 00:00:00	Yogesh	bihar	patna	2018-09
32	B-25610	43	0	3	clothing	saree	2018-09-04 00:00:00	Yogesh	bihar	patna	2018-09
33	B-25610	30	-5	2	furniture	furnishings	2018-09-04 00:00:00	Yogesh	bihar	patna	2018-09
34	B-25611	160	-59	2	clothing	saree	2018-11-04 00:00:00	Anita	kerala	thiruvanantha	2018-11
35	B-25612	259	-55	2	furniture	chairs	2018-12-04 00:00:00	Shrichand	punjab	chandigarh	2018-12
36	B-25613	1603	0	9	clothing	saree	2018-12-04 00:00:00	Mukesh	haryana	chandigarh	2018-12
37	B-25645	273	-87	4	electronics	phones	2018-01-05 00:00:00	Yaanvi	madhya prades	indore	2018-01
38	B-25645	86	0	4	clothing	t-shirt	2018-01-05 00:00:00	Yaanvi	madhya prades	indore	2018-01
39	B-25645	133	-42	1	electronics	printers	2018-01-05 00:00:00	Yaanvi	madhya prades	indore	2018-01
40	B-25645	183	-66	5	electronics	phones	2018-01-05 00:00:00	Yaanvi	madhya prades	indore	2018-01

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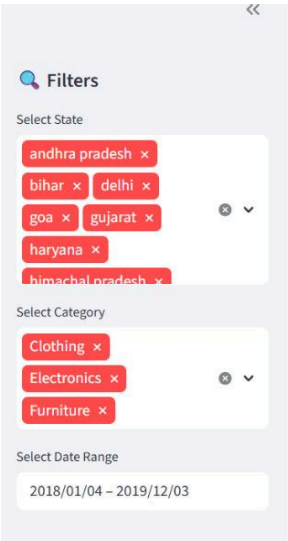
7. Dashboard Components

7.1 Filters

The dashboard includes:

- State filter
- Category filter
- Date range filter

These filters allow dynamic exploration of data.



7.2 Key Performance Indicators (KPIs)

The dashboard displays:

- Total Orders
- Total Sales Amount
- Total Profit
- Total Quantity

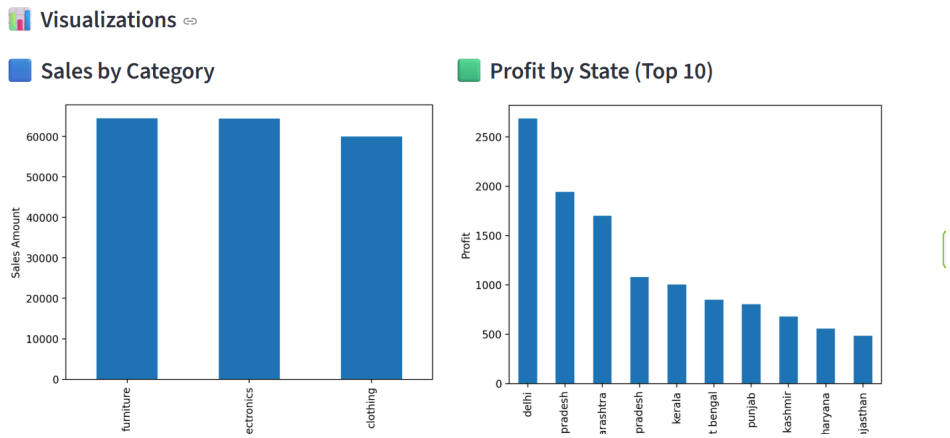
These KPIs provide a quick summary of business performance.

7.3 Visualizations

The following visualizations are included:

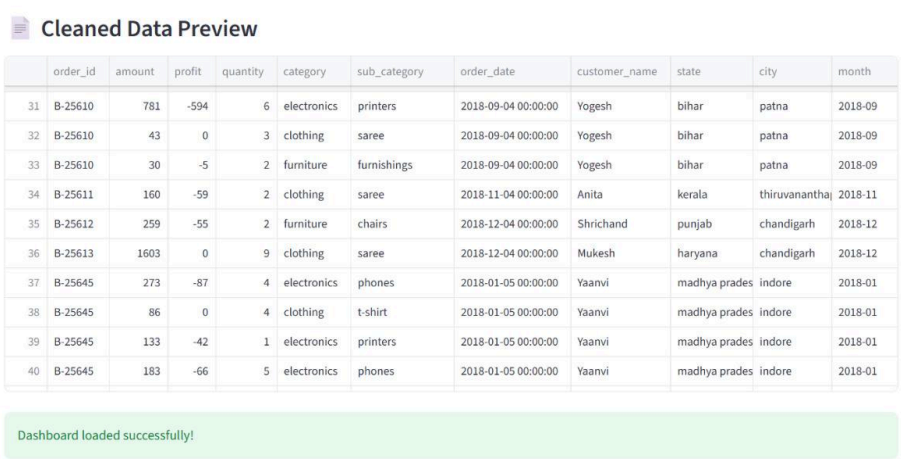
- Sales by Category (Bar Chart)
- Profit by State (Top 10)
- Monthly Sales Trend (Line Chart)
- Top Customers by Sales
- Target vs Actual Sales Comparison

These charts help in identifying trends and patterns.



7.4 Data Preview

- Displays cleaned dataset in tabular format
- Helps verify data correctness



Cleaned Data Preview

	order_id	amount	profit	quantity	category	sub_category	order_date	customer_name	state	city	month
31	B-25610	781	-594	6	electronics	printers	2018-09-04 00:00:00	Yogesh	bihar	patna	2018-09
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Dashboard loaded successfully!

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8. Data Validation

To ensure accuracy:

- Dashboard data was compared with SQLite database
- Verified records and values
- Ensured consistency with ETL outputs

9. Data Relationships

The dataset follows a **one-to-many relationship**:

- One order can contain multiple products
- Customer names may repeat across rows
- This reflects correct relational database design

10. Testing

The system was tested for:

- Correct data loading
- Proper filter functionality
- Accurate KPI calculations
- Error-free visualization rendering

All components worked as expected.

11. Deployment Readiness

The project is structured for easy deployment.

Files Included:

- `app.py`
- `requirements.txt`
- `infosys.db`

Steps to Run:

1. Install dependencies
2. Run the command:

   ```
   streamlit run app.py
   ```

3. Open the dashboard in a browser

12. Conclusion

In this milestone, a complete integration of the ETL pipeline and dashboard visualization was achieved.

- The system enables interactive data analysis
- Provides meaningful insights using visualizations
- Demonstrates a real-world data engineering workflow
- Fully prepared for deployment